1. Take the data you tokenized (your language data) in Assignment-1 and build 4 language model as the following:
    a. Unigram Model
    b. Bigram Model
    c. Trigram Model
    d. Quadrigram Model

Take at least 1000000 sentences and split them into training, development, and test sets. Development and test sets have 1000 sentences each, and training data contains rest of the sentences.

2. Apply Add one/Laplace Smoothing

Use the following formula for the Add one/Laplace Smoothing

$$P_{Laplace}(w_n \mid w_1, w_2, \ldots, w_{n-1}) = \frac{C(w_1, w_2, \ldots, w_n) + 1}{C(w_1, w_2, \ldots, w_{n-1}) + V}$$